

BARSAT KHADKA

CONTACT

Hattiesburg, Mississippi, USA
Barsat.Khadka@usm.edu
+1 (210) 739-3050

ONLINE

www.barsat.dev
[LinkedIn](#) · [GitHub](#)

EDUCATION

The University of Southern Mississippi
Hattiesburg, MS
B.S., Computer Engineering GPA: 3.92

2024 – PRESENT

FELLOWSHIPS & PROGRAMS

Break Through Tech AI/ML

2026 – 2027

Selected participant, [Cornell Tech] — applied machine learning fellowship.

SanDiego Supercomputer Center CIML

JUNE 2026

Summer Institute, UC San Diego — selected for funded HPC/ML training institute.

UR2PHD - Computing Research Association

JULY 2026

Undergrad Research to Phd Program.

PUBLICATIONS

CTS-Bench: Benchmarking Graph Coarsening Trade-offs for GNNs in Clock Tree Synthesis

Khadka, B., Roxy, K., Ahmed, M.R.

ACCEPTED, MLBench Workshop @ ASPLOS 2026

arxiv.org/pdf/2602.19330

PREPRINTS & UNDER REVIEW

SwiftCTS: Fast Cross-Design Prediction and Pareto Optimization of Clock Tree Metrics via Few-Shot Calibration

Khadka, B., Ahmed, M.R.

UNDER REVIEW, ICCAD 2026

arxiv.org/pdf/2606.11348

Filter then Verify: A Multiphase Social Engineering Attack Detection Framework using GNN and BERT for email networks

Khadka, B., Koirala, P., Neupane, K., Rahimi, N.

UNDER REVIEW, Springer

arxiv.org/pdf/2605.17201

MechRL: Reinforcement Learning Agents Perform Circuit Discovery for Mechanistic Interpretability

Khadka, B. (single author)

UNDER REVIEW, TMLR

arxiv.org/pdf/2605.26343

RESEARCH EXPERIENCE

Machine Learning for Chip Design

2025 – PRESENT

Student Researcher

ADVISOR: Dr. Rubel Ahmed, Louisiana Tech (Remote)

- Built CTS-Bench, a benchmark suite evaluating graph-coarsening trade-offs in graph neural networks applied to VLSI clock tree synthesis.
- Designed SwiftCTS, a physics-informed gradient-boosted surrogate (XGBoost/LightGBM) that trains in under 5 s on CPU with sub-millisecond inference; a K-shot calibration scheme cuts power error from 24.5% to 3.3% and wirelength error from 56.6% to under 1% on out-of-distribution macros.

- Led experimental design, ablation studies, and manuscript preparation.

Machine Learning for Cybersecurity

AUG. 2025 – PRESENT

Cyber Innovation Lab, USM

ADVISOR: Dr. Nick Rahimi

- Proposed a two-stage “filter-then-verify” framework using inductive Graph Neural Networks (GraphSAGE) and ModernBERT to detect multi-stage social engineering attacks.
- Built a three-phase structural anomaly pipeline (volume, relational, and contextual) to flag anomalous sender–receiver patterns in dynamic email networks.
- Reached 92% precision by adding a co-attention ModernBERT verifier for semantic intent, cutting false positives on the Enron dataset.

Research Software Engineer

NOV. 2025 – PRESENT

EDT (NOAA Research Agent)

Institute for Advanced Analytics & Society, USM

- Built EDT, an “analyst-at-scale” research agent that quantifies and communicates the downstream impact of NOAA oceanographic and environmental data products on public-sector decision-making.
- Engineered a multi-provider model backend (Claude Sonnet 4.6 / GPT-4.5) with Model Context Protocol (MCP) tools querying federal and academic databases, all running inside single-tenant Daytona sandboxes.
- Implemented deterministic skill packs (citation resolution, reference merging, REACH impact scoring, multi-agent research orchestration) so quantitative steps run as real Node.js/Python rather than LLM math.

INDUSTRY EXPERIENCE

Software Engineering Intern (Backend)

JUN. 2025 – AUG. 2025

Sports Media Inc (Dakdan Worldwide) Dover, DE

- Built Spring Boot microservices and REST APIs to support real-time ad-bidding workflows.
- Integrated Apache Kafka to power live inventory updates across 100+ venues.
- Contributed to CI/CD pipelines using Jenkins, Maven, and Git; deployed services on AWS and GCP, and wrote JUnit test suites in a Unix/Linux environment.

HONORS & SCHOLARSHIPS

Academic Excellence Award — Full Tuition (4 years)

AUG. 2024

Freshman Designation Valedictorian Scholarship

AUG. 2024

Honors Scholar Recognition

AUG. 2024

1st Runner-Up, VOXO Hackathon

NOV. 2024

Reclaiming Time with Generative AI

President’s List

SPRING 2025

SPECIAL PROJECT

Vinaya Journal

2024 – PRESENT

Spring Boot, FastAPI, React, SQLite, Chroma, Electron.js

[Live](#) · [GitHub](#)

- Built a privacy-first, full-stack desktop journaling app integrating a local LLM +

RAG pipeline for offline AI journaling.

- Implemented RESTful Spring Boot endpoints with embedded SQLite, plus a FastAPI microservice with Chroma for semantic/vector search, LLM interactions via Ollama, and mood classification.
- **35+ active users; 150+ GitHub stars**; available for Windows and Linux.

TALKS & TUTORIALS

Positive Uses, Negative Uses, and Limitations of Virtual Reality 2024
MAS 89th Annual Conference, Biloxi, MS

- Presented findings on VR technology and VRET applications.
- Collaborated with peers Anish Kharal and Nhoojah Maharjan in CSC 309; supported by Dr. Sengupta.

TECHNICAL SKILLS

PROGRAMMING: Python, Java, JavaScript/TypeScript, SQL, C++

ML / RESEARCH: PyTorch, HuggingFace Transformers, PEFT, WeightWatcher, LM-Eval-Harness, NumPy, pandas, Chroma

FRAMEWORKS: React, Spring Boot, FastAPI, Maven, Tailwind CSS, PostgreSQL

TOOLS & DEVOPS: Git, GitHub, Docker, AWS, GCP, Apache Kafka, Jenkins, JUnit, Postman

LANGUAGES: English (fluent), Nepali (native), Hindi (advanced)

RESEARCH INTERESTS

Reinforcement Learning, Mechanistic interpretability, Physical Design, World Models, AI for Physical Design, ML in Cybersecurity